DBMS – SALES AND CUSTOMER ANALYTICS SYSTEM

Project Title: Meesho Women's Fashion E-Commerce Order Management System

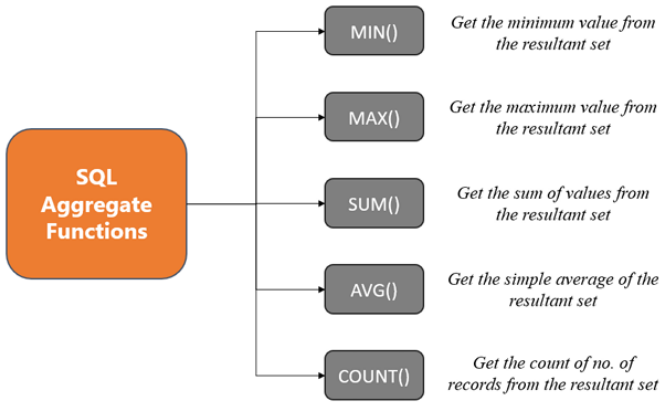
Objectives

1. Apply COUNT(), SUM(), AVG(), MIN(), and MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

1. Aggregate Functions

Definition

Aggregate functions are SQL functions used to perform calculations on a group of rows and return a single result.



1.1 COUNT()

Find the total number of orders:

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          11
```

1.2 SUM()

Find the total sales amount:

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS;

TOTAL_SALES
-----------
      11139
```

1.3 AVG()

Find the average order amount:

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_AMOUNT
  2  FROM ORDERS;

AVERAGE_ORDER_AMOUNT
--------------------
         1012.63636
```

1.4 MIN()

Find the lowest order amount:

```
SQL> SELECT MIN(TOTAL_AMOUNT) AS LOWEST_ORDER_AMOUNT
  2  FROM ORDERS;

LOWEST_ORDER_AMOUNT
-------------------
               599
```

1.5 MAX()

Find the highest order amount:

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_AMOUNT
  2  FROM ORDERS;

HIGHEST_ORDER_AMOUNT
--------------------
                1599
```

2. Generate Total Sales Reports

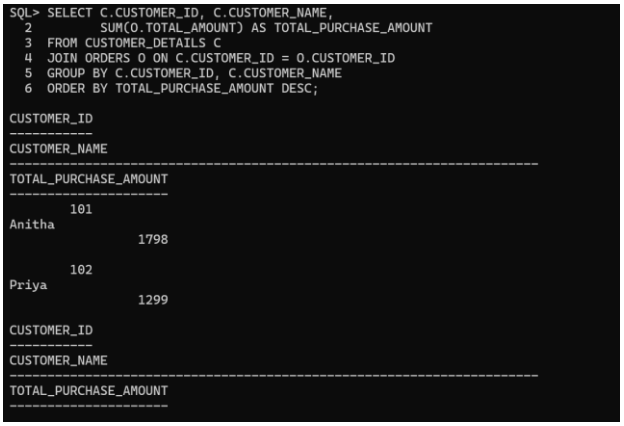
A sales report can be generated using COUNT(), SUM(), and AVG() functions on the ORDERS table.

```
SQL> SELECT
  2      COUNT(ORDER_ID) AS TOTAL_ORDERS,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  4      AVG(TOTAL_AMOUNT) AS AVERAGE_SALES
  5  FROM ORDERS;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES
------------ ----------- -------------
          11       11139    1012.63636
```

2.1 Customer-wise Sales Report

This report displays each customer's total purchase amount and arranges customers according to their purchase amount.



```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  3  FROM CUSTOMER_DETAILS C
  4  JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY TOTAL_PURCHASE_AMOUNT DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
         101
Anitha
                1798

         102
Priya
                1299

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
TOTAL_PURCHASE_AMOUNT
---------------------
```

3. Find Top Customers Based on Purchase Amount

The SUM() function is combined with GROUP BY and ORDER BY to identify customers with higher purchase amounts.

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  3  FROM CUSTOMER_DETAILS C
  4  JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
TOTAL_PURCHASE
--------------
         101
Anitha
           1798

         102
Priya
           1299

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
TOTAL_PURCHASE
--------------
```

The customer appearing at the top has the highest purchase amount in the generated report.

4. Identify Best-Selling Products

The ORDER_DETAILS table contains the quantity of each product ordered. SUM(QUANTITY) finds the total quantity sold for each product.

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         COALESCE(SUM(OD.QUANTITY), 0) AS TOTAL_QUANTITY_SOLD
  4  FROM MEESHO_PRODUCT P
  5  LEFT JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  7  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
       107
Party Wear Gown
                 0

       108
Designer Kurti
                 0

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------

       109
```

This query connects MEESHO_PRODUCT and ORDER_DETAILS, groups records by product, calculates total quantity sold, and displays products from highest to lowest quantity sold.

4.1 Product Sales Amount

This query calculates both the total quantity sold and the sales amount generated by each product.

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         COALESCE(SUM(OD.QUANTITY), 0) AS TOTAL_QUANTITY,
  4         COALESCE(SUM(P.PRICE * OD.QUANTITY), 0) AS TOTAL_SALES_AMOUNT
  5  FROM MEESHO_PRODUCT P
  6  LEFT JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_SALES_AMOUNT DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES_AMOUNT
-------------- ------------------
       107
Party Wear Gown
             0                  0

       108
Designer Kurti
             0                  0

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
TOTAL_QUANTITY TOTAL_SALES_AMOUNT
-------------- ------------------
```

5. Category-wise Sales Analysis

Category-wise analysis helps determine the sales generated by each product category.

```
SQL> SELECT C.CATEGORY_ID, C.CATEGORY_NAME,
  2         SUM(OD.QUANTITY) AS TOTAL_QUANTITY_SOLD,
  3         SUM(P.PRICE * OD.QUANTITY) AS CATEGORY_TOTAL_SALES
  4  FROM CATEGORY C
  5  JOIN MEESHO_PRODUCT P ON C.CATEGORY_ID = P.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY C.CATEGORY_ID, C.CATEGORY_NAME
  8  ORDER BY CATEGORY_TOTAL_SALES DESC;

CATEGORY_ID CATEGORY_NAME
----------- --------------------------------------------------
TOTAL_QUANTITY_SOLD CATEGORY_TOTAL_SALES
------------------- --------------------
          1 Electronics
                  3                1497
```

6. Combined Business Sales Report

A complete sales report combines customer, order, product, and category information.

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID,
  2         P.PRODUCT_NAME, CAT.CATEGORY_NAME,
  3         OD.QUANTITY, P.PRICE,
  4         (OD.QUANTITY * P.PRICE) AS TOTAL_AMOUNT
  5  FROM CUSTOMER_DETAILS C
  6  JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
  8  JOIN MEESHO_PRODUCT P ON OD.PRODUCT_ID = P.PRODUCT_ID
  9  JOIN CATEGORY CAT ON P.CATEGORY_ID = CAT.CATEGORY_ID
 10  ORDER BY O.ORDER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
CATEGORY_NAME                                      QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------
         101

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID
----------
```